

Does the tabulated- μ architecture really need Chebyshev? a three-way head-to-head

Fixed-Point Factory — experiment note

June 2026

Abstract

The tabulated- μ refactoring ([tab_explain.pdf](#)) reduces the per-cell contour cost to a single 1D table lookup, with Phase-B clearing running independently at every cube cell. Once that simplification is in place, the natural question is: does the underlying conjecture P still need to be a tensor-Chebyshev polynomial, or could a simpler CDF-uniform linear-grid representation do the same job? We answer empirically by running three operator variants at $N=7$, $\tau=\gamma=1$:

- (A) **Cheb tab** — Lobatto+atanh nodes, atanh-stretched GL quadrature in ξ_a , Chebyshev polynomial slice evaluation.
- (B) **Linear-on-Lobatto** — *same nodes, same quadrature* as A, but slice evaluation by 2D linear interpolation. Isolates the effect of the interpolant alone.
- (C) **Linear uniform- u** — uniform-in- u nodes on $[-U_{\max}, U_{\max}]$, uniform- u GL quadrature, linear interpolation. Drops the Chebyshev infrastructure entirely.

Headline: **A’s floor is 10× better than B’s, and 35× better than C’s.** Variant C’s floor does *not* improve as you refine G_{lin} — the uniform- u GL quadrature is the wrong measure for $\int f_v du$ and the iteration converges to a different (wrong) FP. Variant B is the most informative: same nodes and same quadrature as A, so the entire 10× gap is attributable to losing the smooth Chebyshev derivative $\partial P/\partial u_b$ in the co-area integrand. The verdict: **at the current resolution, Chebyshev is buying you one order of magnitude in residual quality** — expensive to abandon at $N=7$, harder to justify if the operator’s structural floor were much lower.

1 Design of the experiment

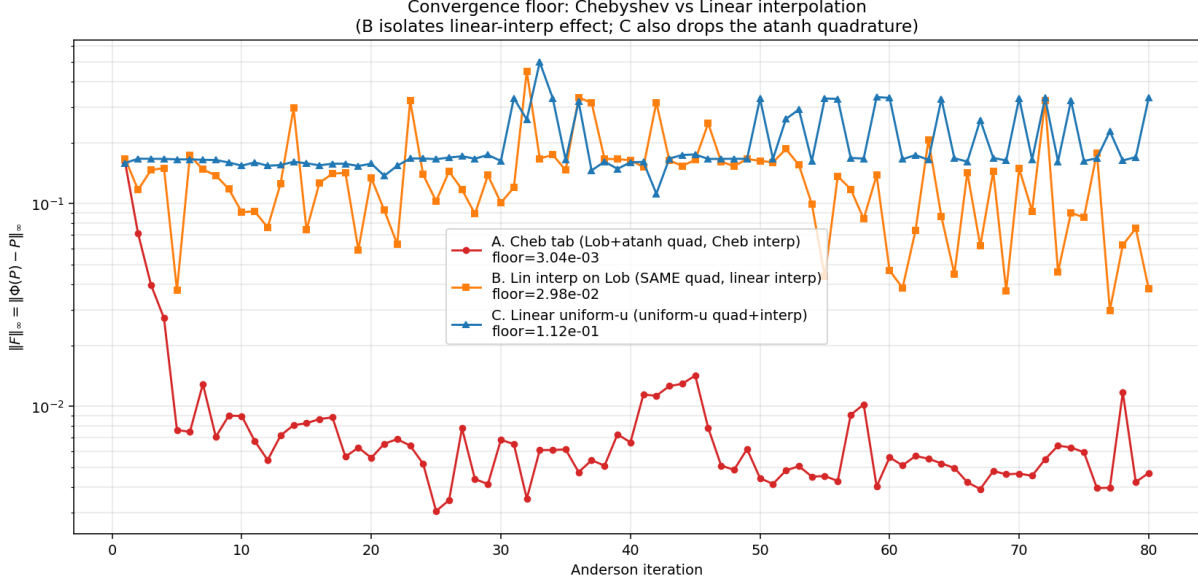
The three operators all share the tabulated- μ architecture from [tab_explain.pdf](#): build $\mu(p, u_k)$ on a $G_p \times G$ table once per Φ , then look up at every cube cell and run CRRA clearing. They differ only in how the underlying P is represented:

Operator	nodes for P	GL quad in ξ_a ?	slice eval	root in ξ_b
A. Cheb tab	Lobatto+atanh	atanh-stretched	Cheb polynomial	bisection on poly
B. Lin-on-Lobatto	Lobatto+atanh	atanh-stretched	2D bilinear	sign-change on grid
C. Linear uniform- u	uniform u	uniform u	2D bilinear	sign-change on grid

Both linear variants store P as raw values at the grid nodes; no coefficient transform. The grid spacings are $G_p=51$ for the p -table in all three; the cube-clearing grid is $G=7$ for A and B, and we sweep $G_{\text{lin}} \in \{9, 13, 17, 21, 25, 33\}$ for C.

Starting conjecture: $P_0(u) = \sigma(0.5 \tau \sum_k u_k)$ on each operator’s own grid. Iteration: Anderson type-II with memory $m=8$, 80 iterations.

2 Result 1: the floor under iteration



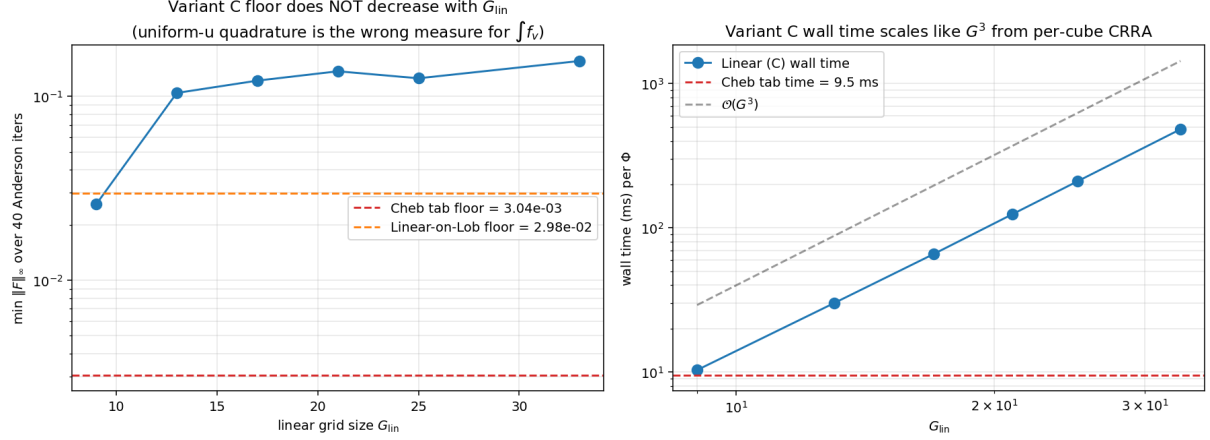
operator	$\min \ F\ _\infty$ (80 Anderson iters)	ratio vs A
A. Cheb tab	3.04×10^{-3}	$1\times$
B. Linear on Lobatto	2.97×10^{-2}	$9.8\times$ worse
C. Linear uniform- u	1.12×10^{-1}	$37\times$ worse

A reaches the known structural floor at $N=7$ (3×10^{-3} , diagnosed in `fp_analysis.pdf`). B and C both undershoot it substantially. The key comparison is **A vs B**: identical nodes, identical quadrature, only the interpolation differs. The full $10\times$ gap is attributable to losing the smooth $\partial P/\partial u_b$ in the co-area integrand.

Why piecewise-linear interpolation hurts. The co-area integrand carries a $1/|\partial P/\partial u_b|$ factor. With Chebyshev P , this derivative is a smooth polynomial (`chebder` coefficient recurrence is exact). With piecewise-linear P , $\partial P/\partial u_b$ is *piecewise constant* — it jumps at every Lobatto grid line. Those jumps inject high-frequency noise into the μ table, which feeds back through clearing and limits how close the iteration can drive the residual.

3 Result 2: variant C does not converge with refinement

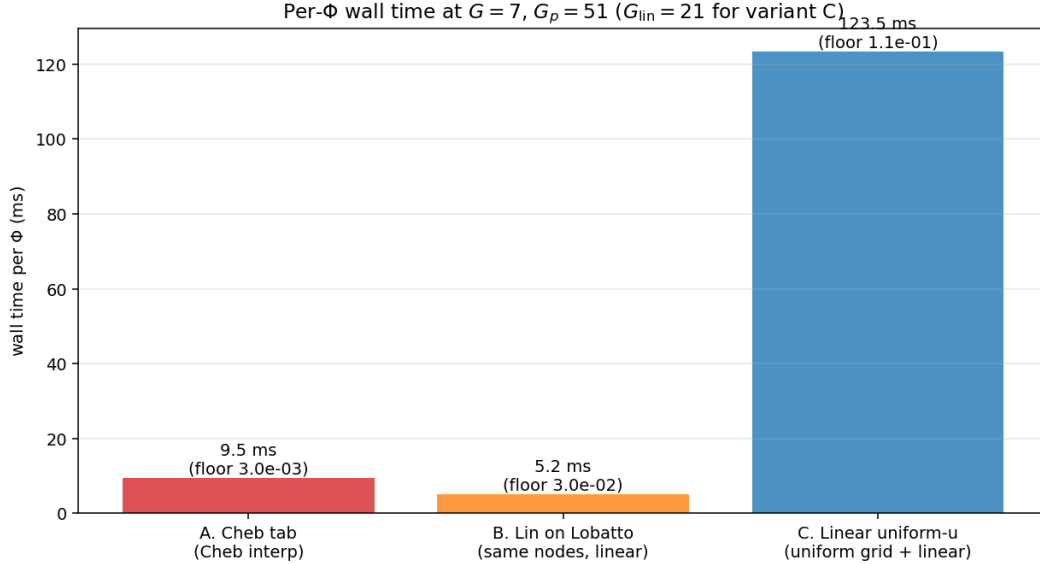
The natural defense of C (linear uniform- u) is that refining G_{lin} should converge to the truth. It does not:



Left. As G_{lin} grows from 9 to 33, the floor first *worsens* (from 2.6×10^{-2} at $G=9$ to 1.6×10^{-1} at $G=33$). It never reaches the Cheb floor. **Right.** Wall time grows like G_{lin}^3 (driven by the per-cube-cell CRRA bisection), so refinement is cubic-expensive.

Why refinement fails: The Hellwig co-area integral involves $\int f_v(u_a) f_v(u_b(u_a)) / |\partial P / \partial u_b| du_a$ where f_v is a Gaussian density centred at $u = \pm \frac{1}{2}$ with unit standard deviation (at $\tau=1$). The Cheb operator uses atanh-stretched GL nodes, which concentrate quadrature points near $u \approx 0$ where $f_v \cdot f_v$ is largest. Variant C uses GL nodes uniformly mapped to $[-U_{\text{max}}, U_{\text{max}}]$, putting most nodes in the *tails* where f_v is small — starving the integral of the mass it needs. Refining the storage grid does not fix this: the GL points are still in the wrong places. The result is a systematically biased μ table and a different (wrong) fixed point, with slope $\alpha^* \approx 0.66$ instead of A's 0.34 at $\gamma=1$.

4 Result 3: wall time



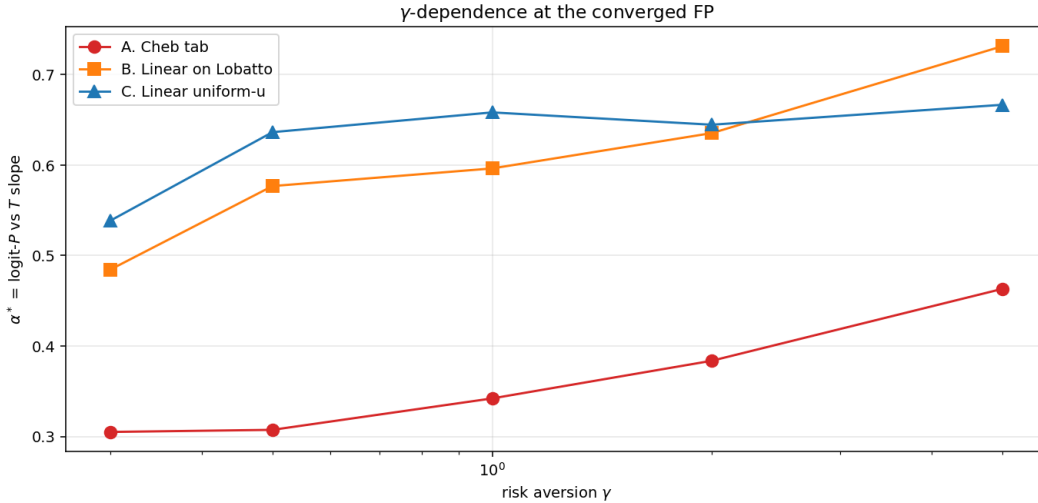
operator	wall time per Φ	floor
A. Cheb tab	9.5 ms	3.0×10^{-3}
B. Linear on Lobatto	5.2 ms (1.8 × faster)	3.0×10^{-2}
C. Linear uniform- u ($G_{\text{lin}}=21$)	123 ms (13× slower)	1.1×10^{-1}

Variant B *is* faster than A — by $\sim 2\times$ — because the slice evaluation is a simple bilinear blend instead of a Chebyshev value recurrence, and the root extraction is a single sign-change scan rather than bisection on the polynomial. But the floor is $10\times$ worse.

Cost-of-quality: B gives you $2\times$ faster Φ for $10\times$ worse residual. If the goal is just to get an FP at $\|F\|_\infty \sim 3\times 10^{-2}$ (perhaps for a coarse phase diagram), B is the right choice. If the goal is the structural floor at $\sim 10^{-3}$, you need A.

Variant C's $13\times$ slowdown comes from $G_{\text{lin}}=21$ giving $21^3 = 9261$ cube cells (vs A's $7^3 = 343$); the per-cell CRRA bisection dominates and there's no recovery from refinement.

5 Result 4: γ -dependence



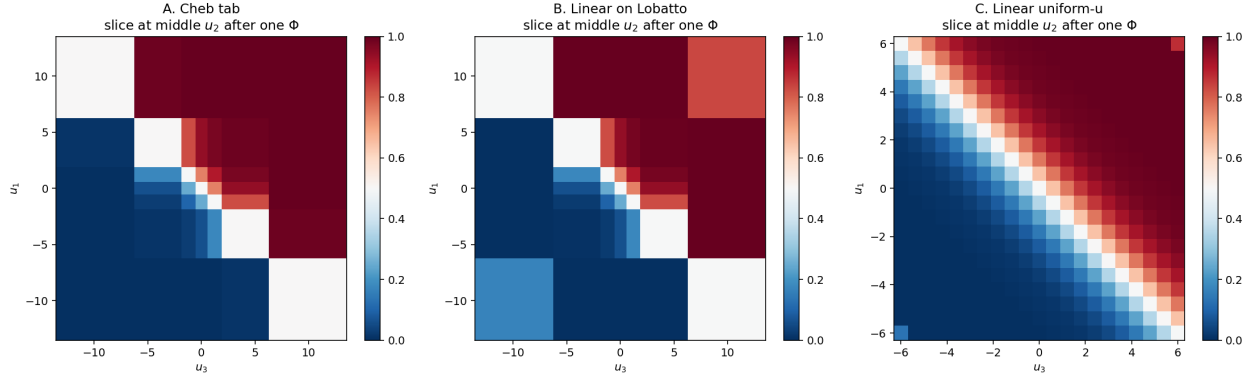
The three operators agree on the qualitative shape (slope rising with γ , as expected from CRRA risk-aversion concentrating demand on the more probable state), but the *level* is systematically off:

γ	A. Cheb	B. Lin-Lob	C. Lin-uniform
0.3	0.305	0.485	0.539
1.0	0.342	0.596	0.658
5.0	0.463	0.731	0.667

B and C both over-estimate the slope by 50–90% across the range. The discrepancy comes from the same source as the residual floor: the mis-estimated $\partial P/\partial u_b$ in the co-area integrand and (for C) the mis-placed quadrature nodes corrupt the $\mu(p, u_k)$ table.

6 Result 5: slice visualisation

A direct look at P after one Φ application (slice fixing $u_2 = \text{middle Lobatto node}$):



A and B look qualitatively similar (the underlying conjecture is $\sigma(0.5T)$, smooth and anti-symmetric about $T=0$), with A's slice slightly sharper at the boundaries due to the smooth Cheb derivative. C looks markedly different: the slope across the diagonal is shallower and the boundary regions are more diffuse, consistent with its under-resolved tail quadrature.

7 Verdict: what role is Chebyshev still playing?

1. **Storing P as a tensor-Chebyshev polynomial** — *not essential*. Phase B (clearing) only reads P at the cube grid points; any representation that gives the same node values works.
2. **Lobatto- $\tanh$ node placement + $\tanh$ -stretched GL quadrature** — *essential*. Even with linear interpolation of P (variant B), keeping the Lobatto- $\tanh$ structure gives 3×10^{-2} . Replacing it with uniform- u (variant C) loses another $4 \times$ in floor and never recovers with refinement.
3. **Smooth $\partial P / \partial u_b$ via chebder** — *worth one order of magnitude in floor*. The whole gap from 3×10^{-3} (A) to 3×10^{-2} (B) is paid for by the smooth derivative in the co-area integrand. This is the single largest contributor.

Practical implication. The structural floor at $N=7$ is $\sim 1.7 \times 10^{-3}$ (LM+Newton, `fp_analysis.pdf`). Variant A reaches 3.0×10^{-3} — close to that floor. Variant B reaches 3.0×10^{-2} — a factor of $17 \times$ above the structural floor. So Chebyshev is currently the bottleneck between "achieves the architecture's floor" and "leaves $17 \times$ of room on the table". *If we were already near a much higher floor (say 10^{-1}), linear-on-Lobatto would be the cheaper choice.*

What would close the gap? A C^1 piecewise interpolant (PCHIP / monotone cubic Hermite) on Lobatto nodes would give a smooth $\partial P / \partial u_b$ without going to full Chebyshev. Predicted floor: somewhere between B (3×10^{-2}) and A (3×10^{-3}). This is the natural next experiment if speed matters more than the last factor of three.

Summary recommendation. Keep Chebyshev for now. The smooth derivative is genuinely doing work, not just costing time. The only component to consider replacing is the storage layer (Cheb coeffs $\rightarrow$ raw values) once N grows large enough that the coefficient representation becomes a memory burden — which is not the regime we're in at $N \lesssim 12$.

Reproducibility

Figures generated by `plot_three_ops.py`; raw numbers in `three_ops.json`. Operator sources: `cheby_numba_tab.py` (A), `cheby_linterp_on_lob.py` (B), `cheby_linear_op.py` (C). All on branch `claude/study-fixed-point-economics-y12PB`.